BUS123 / TEST STUDY SHEET / M03

Trade, chain & cash discounts

Complement / percent paid	`=1-d` 30% off: 1 - 30% = 70% paid.
Discount amount	`=ListPrice*d` $189.99 x 30% = $57.00 saved.
Net price: one discount	`=ListPrice*(1-d)` $189.99 x 70% = $132.99 paid.
Net price: chain discounts	`=ListPrice*(1-d1)*(1-d2)*(1-d3)` $64 x 80% x 90% x 95% = $43.78.
Single equivalent discount	`=1-(1-d1)*(1-d2)*(1-d3)` 20/10/5: 1 - 0.684 = 31.6%. Also =1-NetPrice/ListPrice.
Cash discount savings	`=Invoice*CashRate` $4,200 x 2% = $84 saved if payment qualifies.
Qualifying cash payment	`=Invoice*(1-CashRate)` $4,200 x 98% = $4,116 paid.
Test eligibility with IF	`=IF(PaymentDay<=DeadlineDay,1,0)` Day 12, deadline 15: result 1 (Yes). Day 16: 0 (No).

CHECK BEFORE YOU FINISH

- 2/10 net 30 means 2% off within 10 days; full payment due by day 30.
- Each chain discount applies to the reduced price. Never add the rates.
- Savings and payment are different outputs. Outside the discount window, pay the full invoice.

Markup & markdown

Markup dollars	`=SellingPrice-Cost` $26.10 - $18 = $8.10.
Markup on cost / on selling price	`=MarkupAmount/Cost | =MarkupAmount/SellingPrice` $8.10 / $18 = 45%; $8.10 / $26.10 = 31.03%.
Selling price: markup on cost	`=Cost*(1+MarkupRate)` $18 x 1.45 = $26.10.
Markdown price / markdown dollars	`=OriginalPrice*(1-MarkdownRate)` $52 x 80% = $41.60. Reduction: $52 x 20% = $10.40.
Sellable units after waste	`=ProducedUnits*(1-WasteRate)` 120 x 85% = 102 sellable boxes.

Target revenue / required price	`=ProducedUnits*CostPerUnit*(1+MarkupRate)`
	`=TargetRevenue/SellableUnits`
	120 x $18 x 1.45 = $3,132; $3,132 / 102 = $30.71 per box.

CHECK BEFORE YOU FINISH

- Waste leaves fewer units to recover the cost of all units produced. Keep precision until the final result.

Payroll & straight-line depreciation

Salary per pay period	`=AnnualSalary/PayPeriods` $52,000 / 52 = $1,000 weekly; / 12 = $4,333.33 monthly.
Regular pay	`=MIN(Hours,Cap)*Rate` 43 hours, 40-hour cap, $44/hour: 40 x $44 = $1,760.

Overtime pay / gross pay	`=MAX(Hours-Cap,0)*Rate*Multiplier`
	`=RegularPay+OvertimePay`
	3 x $44 x 1.5 = $198 overtime; gross = $1,958 before deductions.

Same-schedule team pay / service hours	`=TeamSize*GrossPay | =TeamSize*Hours` 9 x $1,958 = $17,622; 9 x 43 = 387 service hours.

Equal-share alternative: team hours	`=MIN(RequiredHours,TeamSize*Cap)`
	`=MAX(RequiredHours-TeamSize*Cap,0)`
	10 nurses, 387 hours, cap 40: regular = 387, overtime = 0. Average = 387 / 10 = 38.7 hours.

Alternative wages / weekly savings	`=RegularTeamHours*Rate+OTTeamHours*Rate*Multiplier`
	`=CurrentWages-AlternativeWages`
	387 x $44 = $17,028; $17,622 - $17,028 = $594 saved weekly.

Annual savings / avoided OT premium	`=WeeklySavings*52`
	`=OTHours*Rate*(Multiplier-1)`
	$594 x 52 = $30,888 yearly; 27 x $44 x 0.5 = $594 premium avoided.

Annual straight-line depreciation	`=(Cost-ResidualValue)/UsefulLife` ($84,000 - $12,000) / 6 = $12,000 per year.

Accumulated depreciation / book value	`=AnnualExpense*Year`
	`=Cost-AccumulatedDepreciation`
	Year 3: $12,000 x 3 = $36,000 accumulated; $84,000 - $36,000 = $48,000 book value.

CHECK BEFORE YOU FINISH

- MIN caps regular hours. MAX prevents negative overtime. Apply the supplied cap and multiplier.
- Team-hour formulas assume equal sharing under the lesson schedule. Benefits and hiring costs are excluded.
- Book value uses accumulated depreciation. Year 6 ends at $12,000 residual; do not extend past useful life.
- Depreciation is an expense allocation, not a new annual cash payment or market value.

Inventory & COGS

Layer cost / units left / ending value	`=UnitsSold*UnitCost`
	`=AvailableUnits-UnitsSold | =UnitsLeft*UnitCost`
	First FIFO layer: 10 x $16 = $160 cost sold; 10 - 10 = 0 left.

FIFO: assign oldest costs first	`=10*16+20*20` COGS = $560; ending inventory = 20 x $22 = $440.
LIFO: assign newest costs first	`=20*22+10*20` COGS = $640; ending inventory = 10 x $16 + 10 x $20 = $360.
Weighted-average unit cost	`=TotalCostAvailable/TotalUnitsAvailable` $1,000 / 50 = $20 per hoodie. Use quantities as weights.
WAC COGS / ending inventory	`=UnitsSold*WACUnitCost | =UnitsLeft*WACUnitCost` 30 x $20 = $600 COGS; 20 x $20 = $400 ending inventory.

Goods available / COGS	`=BeginningInventory+Purchases`
	`=GoodsAvailable-EndingInventory`
	$160 + $840 = $1,000 available. FIFO: $1,000 - $440 = $560 COGS.

Revenue / gross profit / gross margin	`=UnitsSold*SellingPrice`
	`=Revenue-COGS | =GrossProfit/Revenue`
	Revenue $1,200; FIFO gross profit $640; margin $640 / $1,200 = 53.33%.

Margin versus markup / conversions	`=(Price-Cost)/Price | =(Price-Cost)/Cost`
	`=Markup/(1+Markup) | =Margin/(1-Margin)`
	$40 price, $20 cost: margin 50%, markup 100%. 100% / (1 + 100%) = 50%.

CHECK BEFORE YOU FINISH

- Reconcile units sold + units left = available; COGS + ending inventory = total available cost.
- A balanced total does not prove FIFO or LIFO: check the layer order too.
- Overstated ending inventory understates current COGS and overstates current profit.

Sales, vehicle excise & property taxes

Taxable subtotal	`=Price-TradeDiscount` Tidal order: $480 - $80 = $400 taxable.
Sales tax owed	`=TaxableSubtotal*TaxRate` $400 x 6.25% = $25.
Total customer payment	`=TaxableSubtotal+Tax+Shipping` $400 + $25 + $15 = $440.
Back-calculate pre-tax sales	`=TaxInclusiveReceipts/(1+TaxRate)` $52,000 / 1.0625 = $48,941.18 pre-tax sales. Assumes all receipts subject to the same rate, no shipping.
Collected tax in inclusive receipts	`=TaxInclusiveReceipts-PreTaxSales` $52,000 - $48,941.18 = $3,058.82 collected tax.
Vehicle excise value	`=OriginalListPrice*YearPercent` Meridian car: $32,000 original list x 60% (second year) = $19,200.
Annual vehicle excise	`=ExciseValue/1000*25` $19,200 / 1,000 x $25 = $480 annual excise; full-year example.
Property: thousands of assessed value	`=AssessedValue/1000` Meridian illustrative office: $820,000 / 1,000 = 820.
Annual property tax	`=AssessedValue/1000*LocalRatePer1000` Beverly FY2026 commercial example: 820 x $20.98 = $17,203.60.
Monthly amount to set aside	`=AnnualTax/12` Property: $17,203.60 / 12 = $1,433.63. Vehicle: $480 / 12 = $40.

CHECK BEFORE YOU FINISH

- Vehicle-year percentages: manufacture 90%; second 60%; third 40%; fourth 25%; fifth and later 10%.
- Property rates supplied in M07: commercial $20.98; residential $10.81 per $1,000, FY2026.
- Use assessed value directly for property. Use original manufacturer list price for vehicle excise.
- A rate per $1,000 is a dollar rate, not a percent. Monthly savings are budgeting amounts, not bill frequency.
- Collected sales tax is a liability, not sales revenue. Examples assume no exemptions or abatements.